POWER BI PROJECT

Ecommerce Sales Analysis Dashboard using Power BI

An interactive dashboard to monitor sales performance, profitability, customer behaviour, product categories and regional performance.

Presented by Asrarul Haque

PROJECT OVERVIEW

Objective of the Project

The objective of this project is to analyze ecommerce sales data and provide management with an interactive dashboard to monitor sales performance, profitability, customer behavior, product categories, and regional performance.

The dashboard enables decision-makers to identify sales trends, profitable categories, and areas requiring business improvement.

AT A GLANCE

- **Tool**
  Microsoft Power BI
- **Domain**
  Ecommerce Sales
- **Focus**
  Sales, Profit & Customer Insights
- **Report Type**
  Interactive Dashboard

BUSINESS PROBLEM

Questions the Dashboard Answers

Ecommerce companies generate thousands of sales transactions every month. Without proper reporting, it becomes difficult to answer important business questions.

1. **What are the total sales and profits?**
2. **Which state generates the highest revenue?**
3. **Which category sells the most products?**
4. **Which payment method is most popular?**
5. **Which months are profitable?**
6. **Which sub-category generates the highest profit?**

DATASET

Ecommerce Transaction Data

The dashboard uses ecommerce transaction data. Typical columns include:

Order ID	**Order Date**	**Customer Name**	**State**	**Category**
Sub Category	**Sales Amount**	**Quantity**	**Profit**	**Payment Mode**

This structured dataset forms the foundation for all measures, visuals, and insights built into the dashboard.

DATA PREPARATION

Cleaning & Preparing the Data

1. Removing duplicate records
2. Handling blank values
3. Correcting inconsistent category names
4. Formatting date columns
5. Verifying sales and profit values
6. Creating relationships between tables (if multiple tables were used)

TOOLS & TECHNIQUES

Power BI Features Used

- Power Query Editor
- Data Cleaning
- Data Modeling
- Relationships
- DAX Measures
- Card Visuals
- Bar Charts
- Donut Charts
- Column Charts
- Slicers & Filters
- Dashboard Formatting
- Interactive Visuals

KEY PERFORMANCE INDICATORS

Headline KPI Cards

438K

TOTAL SALES

Displays the total revenue generated.

37K

TOTAL PROFIT

Shows overall business profit.

121K

AVERAGE SALES

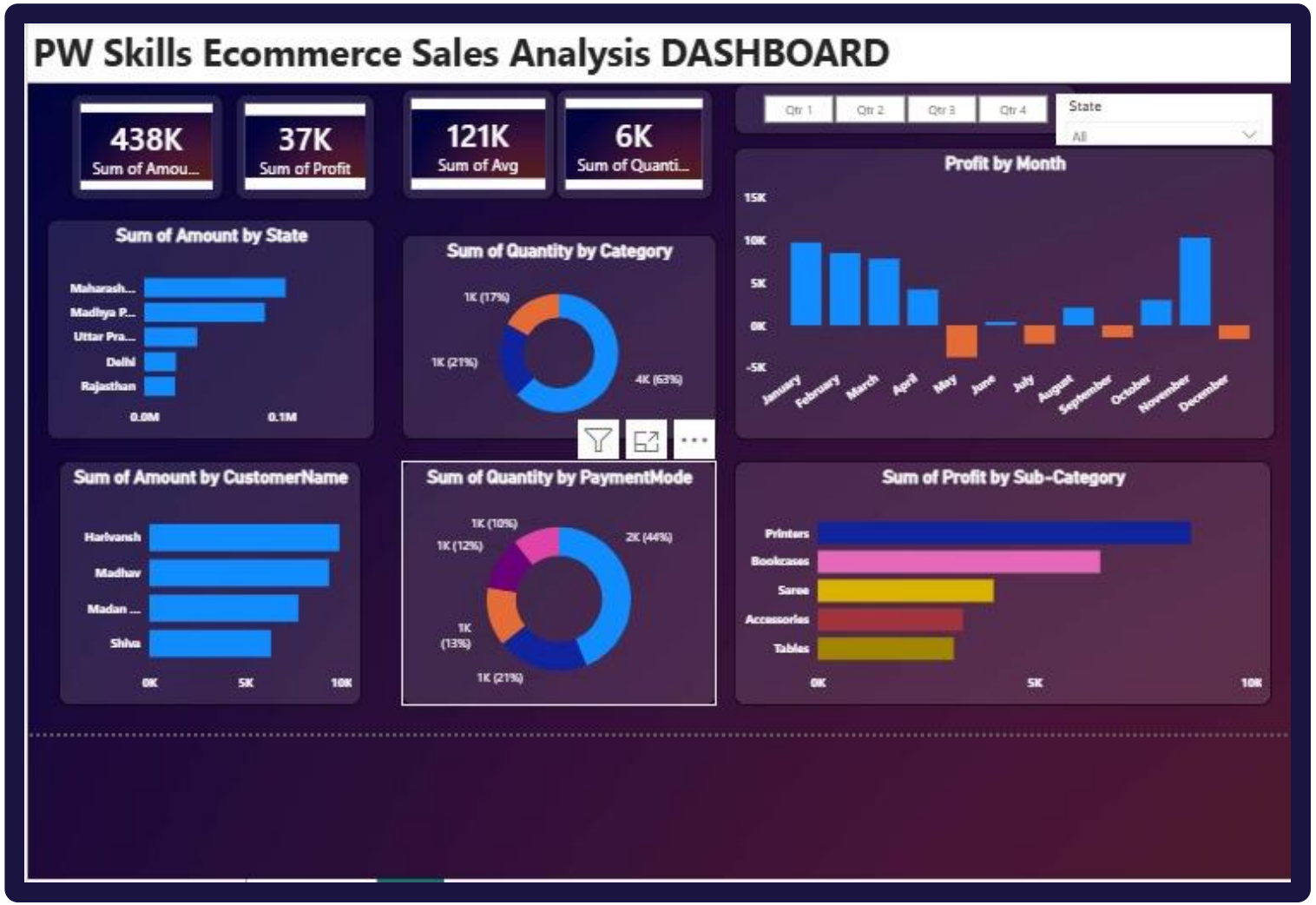
Displays average sales value.

6K

TOTAL QUANTITY

Shows total products sold.

PW Skills Ecommerce Sales Analysis Dashboard



DASHBOARD VISUALIZATIONS

Key Visuals — Part 1

1

Sales by State

Displays total sales generated from each state.

BUSINESS VALUE

Helps identify high-performing and low-performing regions.

2

Quantity by Category

Shows product quantity sold across different categories.

BUSINESS VALUE

Identifies the most demanded product category.

3

Profit by Month

Displays monthly profit trends.

BUSINESS VALUE

Helps identify profitable months and months with losses.

DASHBOARD VISUALIZATIONS

Key Visuals — Part 2

4

Amount by Customer

Shows customers generating the highest sales.

BUSINESS VALUE

Useful for identifying valuable customers.

5

Quantity by Payment Mode

Displays payment method distribution.

BUSINESS VALUE

Helps understand customer payment preferences.

6

Profit by Sub-Category

Shows profit generated by different product sub-categories.

BUSINESS VALUE

Helps identify the most profitable products.

INTERACTIVITY

Dashboard Filters

The dashboard includes the following filters, allowing users to explore the data dynamically:

Q Quarter Filter

- **Quarter 1**
- **Quarter 2**
- **Quarter 3**
- **Quarter 4**

S State Filter

Users can filter the entire dashboard by state, allowing region-specific analysis of sales, profit, and customer behavior in real time.

FINDINGS

Key Insights from the Dashboard

- Total Sales reached 438K.
- Total Profit is 37K.
- Total Quantity Sold is 6K.
- Maharashtra is one of the highest revenue-generating states.
- Printers appear to be the highest profit-generating sub-category.
- Some months show negative profit, indicating seasonal or operational challenges.
- Payment mode analysis helps understand customer buying behavior.

RECOMMENDATIONS

Business Recommendations

1. Increase marketing efforts in high-performing states.
2. Investigate reasons for losses during low-profit months.
3. Promote high-profit sub-categories such as Printers.
4. Improve inventory planning based on category demand.
5. Offer payment options preferred by customers.

REFLECTION

Challenges Faced & Future Improvements

CHALLENGES FACED

- Cleaning raw ecommerce data
- Building relationships between tables
- Writing DAX measures for KPIs
- Selecting appropriate visuals
- Designing an interactive dashboard

FUTURE IMPROVEMENTS

Sales Forecasting	**Customer Segmentation**
RFM Analysis	**Region-wise Profit Map**
Year-over-Year Growth	**Top 10 Products**
Drill-through Pages	**Tooltip Reports**
Bookmarks	**Row-Level Security (RLS)**

LEARNING OUTCOME

Skills Gained from this Project

Through this project, practical experience was gained across the full Power BI workflow — from raw data to a polished, decision-ready dashboard.

Power Query

Importing and cleaning raw ecommerce data.

DAX

Writing measures for Total Sales, Profit, Avg Sales & Quantity.

Data Modeling

Building relationships between tables.

Dashboard Design

Creating an interactive, decision-ready layout.

Business Analysis

Turning raw numbers into actionable insight.

THANK YOU

Questions & Discussion

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